

ONLINE MASTERS IN **DATA SCIENCE**

DSC 257R - UNSUPERVISED LEARNING

DISTANCE-PRESERVING EMBEDDINGS

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Distance-Preserving Embeddings

- **Input:** An $n \times n$ matrix of pairwise distances

D_{ij} = desired distance between points i and j ,

as well as an integer k .

- **Desired output:** An embedding $z^{(1)}, \dots, z^{(n)} \in \mathbb{R}^k$ that realizes these distances as closely as possible, that is, for which

$$\|z^{(i)} - z^{(j)}\| \approx D_{ij}.$$

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Does such an embedding even exist?

Example 1

Is there an embedding $z^{(1)}, z^{(2)}, z^{(3)} \in \mathbb{R}^k$ that realizes these distances?

$$D = \begin{pmatrix} 0 & 3 & 5 \\ 3 & 0 & 4 \\ 5 & 4 & 0 \end{pmatrix}$$

Example 2

Is there an embedding $z^{(1)}, z^{(2)}, z^{(3)}, z^{(4)} \in \mathbb{R}^k$ that realizes these distances?

$$D = \begin{pmatrix} 0 & 1 & 2 & 1 \\ 1 & 0 & 1 & 2 \\ 2 & 1 & 0 & 1 \\ 1 & 2 & 1 & 0 \end{pmatrix}$$